# DEVANSH KANDPAL

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## ACADEMIC DETAILS

| Year | Degree / Board | Institute | GPA / Marks(%) |
|---|---|---|---|
| --- | B.Tech in Production & Industrial Engineering | Indian Institute of Technology Delhi | 9.03 |
| 2024 | CBSE | GVN the Global School | 97% |
| 2022 | CBSE | Delhi Public School Bhopal | 96% |

## SCHOLASTIC ACHIEVEMENTS

- **Department Rank 4:** Based on CGPA at the end of Sem 4, among 80+ students in Production and Industrial Engg.
- **Semester Merit:** Received the Semester Merit Award (Top 7%) for the third semester based on semester GPA.
- **Summer Undergraduate Research Award:** Research proposal selected from 40+ applicants in the CSE department.
- **IMC Prosperity 2026:** Ranked **#146 in India** and **#68 global (manual)** in a trading competition with 18,000 teams.

## INTERNSHIPS

**Cloudflare** · Software Engineer Intern, Bengaluru (May – July, 2026)

- Worked on Statusflare, a **status monitoring service and public status page on the Workers platform** with automatic health checks, New Relic/DataDog integrations, and **email notifications** for incidents and maintenance.
- Built Continual Learning to review AI Gateway logs: clustered conversations by intent with embeddings, wrote a rubric per cluster, and ran automatic evaluations on real conversations, revealing which requests the product missed.
- Implemented **GEPA** to study those failures, revise prompts, and keep the best-scoring versions, and built the dashboard plus a reproducible test bench that saves every prompt version side by side for comparison afterwards.

## PROJECTS

**Lapis** · Self Project · TypeScript, Cloudflare Workers, React (Jun, 2026 – Ongoing)

- Built **Lapis**, a private web vault for Obsidian notes, with browser editing, wikilinks, backlinks, and SQLite FTS5 search, synced both ways by an Obsidian plugin.
- Kept each vault consistent with a **Durable Object** that applies text patches and writes conflict notes, with offline replay, sealed Git history, and MCP access only for opted-in vaults.

**BitTorrent CLI Client** · Self Project · Rust, TCP, Concurrency (May – June, 2026)

- Built a BitTorrent **file-sharing client in Rust with a custom bencode parser for .torrent files and SHA-1 verified piece reassembly.**
- Implemented rarest-first piece selection, tracking availability with **bitfields**; implemented a choke algorithm to prioritize altruistic peers.
- Refactored the sequential actors to **async Tokio tasks with a sliding window of in-flight requests (8 peers × 12 requests)** to overlap network latency across peers, improving throughput from **12.08 MiB/s to 622 MiB/s (51.5×)**.

**Jamming Attacks on RAG Systems** · SURA, Prof Vireshwar Kumar · RAG, Black-box optimization (Apr, 2026 – Ongoing)

- Extended the single-query jamming attack on Retrieval-Augmented Generation (RAG) systems to wider class attacks.
- Characterised a **geometric bound on single-document attacks**, proving the bound is structural to the query embedding geometry.
- Quantified multi-document attack complexity via a clean set-cover formulation, showing that the number of required blocker documents scales linearly with the within-class query diversity and matches PCA-predicted dimensionality.

**TouristGPT: Algorithmic reasoning in small models** · ARIES Freshers Project · PyTorch, TRL (Feb, 2026 – Ongoing)

- Led an effort to **SFT and GRPO** fine tune a small language model (8B) for algorithmic reasoning in C++ at par with frontier models.
- Worked on a pipeline for generating CoT traces that explore more diverse solutions, published on Huggingface.

**FinTerminal: Multi-Agent Portfolio Manager** · Inter IIT Tech · LangGraph, Next.JS, FastAPI, Docker (Oct – Dec, 2025)

- Contributed to a reactive **portfolio management engine**, integrating insights from technical, fundamental, and live news **AI agents**.
- Built the Fundamental Analysis **multi-agent flow in LangGraph** evaluating financial health from SEC filings, including Forensic, Credit, and Equity analysts, achieving **62% directional accuracy** against subsequent-quarter.
- Worked on Omnus, an agent with portfolio and market context + tools, adding a natural interaction layer.

**ChatIITD + Degree Planner** · ARIES, Devclub · Agentic AI, AWS Bedrock, Google OR-Tools (Jun, 2025 – Mar, 2026)

- Created an **agentic chatbot** for academic queries, leveraging data sources on courses, departments and boards.
- Contributed to a **CP-SAT based degree planner** to create course plans, enforcing prerequisites and slot constraints.

## TECHNICAL SKILLS

- **Programming and Systems:** C++, Python, Typescript, Javascript, Rust, SQL, Shell, Linux (arch btw), git, Docker, Nginx
- **Full stack + AI development:** Node.js, fastapi, NestJS, React, TanStack, Flue, LangChain, MCP, API Design, Cloudflare
- **Data analysis and machine learning – Python:** pytorch, scikit-learn, xgboost, pandas, numpy, matplotlib, optuna

## EXTRA CURRICULAR ACTIVITIES

- **Winner, BlackRock HackKnight 2026:** Built an explainable asset trader, placed first among 40 teams, won Rs.90,000.
- **Winner, CyberPeace x Cloudflare Buildathon:** Built a multilingual voice-assisted PDF form filling application.
- **Inter-IIT Tech Meet 14.0:** Worked on the **Pathway: Agentic AI** problem statement | Awarded Best High Prep Team
- **Second Runner Up, Claw & Shield, Cognizance IITR:** Added safe personal information handling to OpenClaw.

## POSITIONS OF RESPONSIBILITY

**Coordinator, ARIES** Organised Tryst events, Shipathon, led freshers' projects, awarded Backbone of ARIES.
**Coordinator, DevClub** Contributed to Projects Portal, ChatIITD & IITD Wiki, IITD contingent member to Cognizance IIT Roorkee.